2. **(a)** State the conditions for a body to be in equilibirum.

1. .....

.....

2. .....

..... [2]

(b) Fig. 2.1 shows a 70 kg climber hanging by only the crimp hold of one hand on the edge of a shallow horizontal ledge in a rock wall. The fingers are pressed down.

Her feet touch the rough rock wall at distance H = 2.0 m directly below her crimped fingers. The frictional force between her feet and the wall is 32 N. Her centre of mass is at distance a = 0.20 m from the wall.

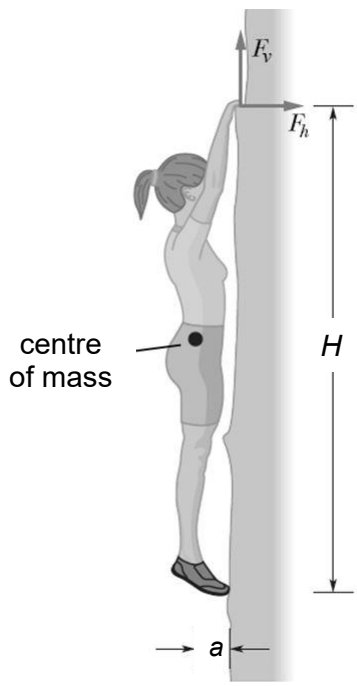

Fig. 2.1

(i) Determine the horizontal frictional force F_h acting on the fingers.

F_h = ... N [2]

(ii) Determine the vertical force F_v acting on the fingers.

F_v = ... N [2]

(iii) The weight W of the climber and the resultant force F exerted by the ledge on her fingers are indicated in Fig. 2.2. Draw and label the force S exerted by the rock wall on her feet.

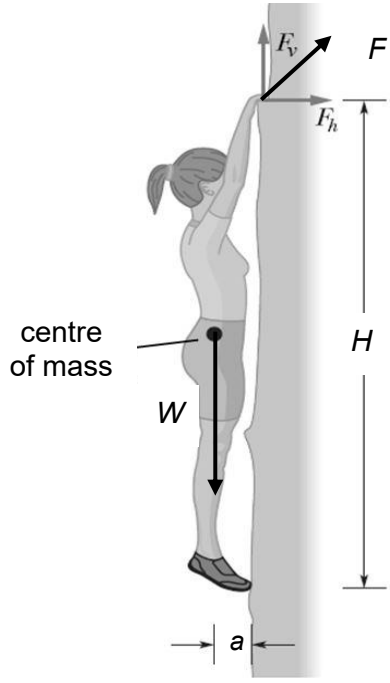

Fig. 2.2

[2]